

Konstantinos Stamatis

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Materials scientist with over five years of experience leading projects, driven by a strong passion for contributing to innovative real-world applications and supporting strategic decision-making. Motivated by a continuous desire to excel in both science and leadership, I aim to combine my engineering expertise with in-depth knowledge of R&D processes to drive innovation and achieve business success.

EDUCATION

Texas A&M University

Expected Graduation: December 2027

Materials Science & Engineering, PhD

Focus on Polymeric Actuators for a Robotic Hand and Artificial Muscles

NCSR Demokritos

Institute of Nanoscience & Nanotechnology, MS

Development of Metal Complexes for Anticancer Drugs

Aristotle University of Thessaloniki

Chemistry Department, BS, Valedictorian

Major in Inorganic Chemistry

RESEARCH EXPERIENCE

NSF HAND Engineering Research Center (ERC) — Texas A&M University - College Station, TX, USA

August 2023- Present

Graduate Research Assistant, Dr. Taylor Ware and Dr. Svetlana Sukhishvili Laboratory,

Materials Science & Engineering and Biomedical Engineering Department

- Design and synthesize soft polymeric actuators — including **Liquid Crystal Elastomers (LCEs)** and **polydiethylsiloxane (PDES)** — for integration into an **AI-driven robotic hand**, as part of the NSF HAND ERC (\$52M, 10-year initiative; 5-university consortium: Texas A&M, Northwestern, CMU, MIT, FAMU).
- Lead cross-laboratory collaboration on actuator design and materials integration across the consortium.
- Perform comprehensive thermomechanical characterization of materials to evaluate and optimize actuation performance
- Built Python scripts using AI-assisted workflows (Claude) to automate DMA/DSC/TGA data analysis and accelerate literature synthesis, cutting manual analysis time.
- Drive commercialization efforts for advanced polymeric actuator materials, targeting applications in robotics and biomedical engineering.
- Mentor undergraduate researchers and serve as a Research Experiences for Undergraduates (REU) mentor.

NCSR Demokritos – Athens, Greece

November 2021-August 2023

Graduate Research Assistant, Dr. Aikaterini Dendrinou-Samara, Dr. Aikaterini Raptopoulou & Dr. Vassilis Psycharis laboratory

- Led a 1-year project across 3 labs and partnered with cross-functional teams to synthesize and characterize four zinc complexes, evaluating their antimicrobial and anticancer properties.
- Developed novel metal complexes demonstrating in vitro antimicrobial and anticancer activities superior to those of commercially available drugs.
- Speaker at the Joint CTMNM/NAGC Conference, May 2023, Spetses, Greece.

NCSR Demokritos – Athens, Greece

September 2021-October 2021

Research Intern, Dr. Margarit Gjoka laboratory

- Conducted research on advanced permanent magnet materials, with a focus on the synthesis and characterization of magnetic materials.
- Presented research findings at the 15th Joint MMM-INTERMAG Conference. January 2022, New Orleans, USA, and at the 1st Conference of NIP, February 2022, Tirana, Albania.

Aristotle University of Thessaloniki – Thessaloniki, Greece

September 2017-July 2021

Undergraduate Research Assistant, Dr. Angaridis Panagiotis Laboratory

- Selected from a competitive pool of students to conduct thesis research with a dissertation.
- Designed and synthesized novel copper complexes for OLED applications, conducting an in-depth study of OLED technology.
- Developed strong laboratory skills in synthesis and characterization techniques.

PUBLICATIONS

- **First author** on a manuscript currently under review at Small, developed with collaborators from Texas A&M University. Title withheld until publication. Accepted at **Small (Impact Factor 12.1)**.

SCHOLARSHIPS AND AWARDS

SPE (Society of Plastics Engineers) International Polyolefins Conference Scholarship

February 2026

- Competitive graduate-level scholarship awarded for outstanding research excellence at Texas A&M University.

Texas Soft Matter, Rice University

August 2025

- Presentation Award Winner selected from 43 student participants for excellence in research presentation.

Gerondelis Foundation Grant

July 2025

- Awarded scholarship supporting Greek nationals in U.S. graduate programs

Henry Kahn Memorial Scholarship	April 2025
<i>Society of Plastics Engineers (SPE) South Texas Section</i>	
Awarded for excellence in academics and research among graduate and undergraduate students across Texas.	
Material Advantage Congressional Visits Day (CVD) Participant	April 2025
- Participated in the 2025 Material Advantage Congressional Visits Day in Washington, D.C. - Awarded a \$320 travel grant to support participation in the event. - Engaged with Texas Senators and Representatives to advocate for the support of education and research in the physical sciences and engineering, emphasizing their role in driving innovation and scientific progress.	
Texas A&M University	November 2024
<i>Aggies Invent: Problems Worth Solving Competition by the Entrepreneurship Department</i>	
- Addressing Costly and Life-Threatening Catheter-Associated UTIs - First Place Award for Designing the Most Innovative Prototype	
LEADERSHIP & SERVICE	
NSF HAND Engineering Research Center (ERC) — Texas A&M University	March 2026 - Present
<i>Officer, Student Leadership Council (SLC)</i>	
Spearhead professional development programming connecting NSF HAND graduate students with industry leaders in robotics and advanced manufacturing	
Texas A&M University	Summer 2025- Present
<i>President, Society of Plastics Engineers (SPE)</i>	
- Global professional organization advancing knowledge and education in the plastics industry - Plan networking events with polymer industry professionals and coordinate site visits to major companies including Dow, Chevron Phillips, and Kuraray.	
Texas A&M University	Spring 2025 & Fall 2025
<i>Graduate Teaching Assistant</i>	
- Teach polymer labs to 80 undergraduate students at the Baker Hughes Materials Lab, Texas A&M University. - Supervise and mentor laboratory students in preparing detailed 30-page reports, executing advanced characterization and assay development protocols, and managing \$200K worth of lab equipment and materials.	
Texas A&M University	Fall 2023-Summer 2025
<i>Vice President, Materials Science & Engineering Graduate Student Advisory Council</i>	
- Collaborated with Materials Science and Engineering department leadership to enhance student life and expand academic resources. - Hosted town halls to engage graduate students, collect feedback, and successfully advocate for improvements that enhanced student experiences.	
Texas A&M University	Fall 2023-Present
<i>Founder & President, Hellenic Student Association</i>	
- Founded, led, and secured official recognition for the organization at Texas A&M University, inspiring student engagement, recruiting faculty support, and fostering active participation. - Collaborated with Halliburton Engineering Global Programs leadership and Greek faculty to expand Texas A&M's international student recruitment in Greece and Cyprus	
Organizations	Spring 2025-Present
<i>Member, Material Advantage at Texas A&M University</i>	
<i>Member, Minerals, Metals & Materials Society (TMS)</i>	
<i>Member, American Society for Metals (ASM)</i>	
<i>Member, American Ceramic Society (ACerS)</i>	
<i>Member, Association for Iron & Steel Technology (AIST)</i>	
Erasmus+	October 2018
<i>Greek Representative in Berlin, Germany, European Youth Exchange Program</i>	
- Selected and fully funded by the United Societies of the Balkans, a non-profit organization (NGO). - Promoted youth mobility and collaborated with teams from five different countries: Bulgaria, Poland, Spain, Romania, and Italy. - Addressed environmental awareness issues in European countries and explored renewable energy sources.	
Aristotle University of Thessaloniki	September 2018- August 2019
<i>Social Manager, Chemistry Professional Organization "ReAcTiON"</i>	
Organize events and outreach programs to promote the field of chemistry to the public.	
TECHNICAL SKILLS	
Material characterization: Dynamic Mechanical Analysis (DMA), Universal Testing Machine (UTM), Differential Scanning Calorimetry (DSC), Gel Permeation Chromatography (GPC), Infrared (IR) Spectroscopy, Scanning Electron Microscopy (SEM), Thermogravimetric Analysis (TGA), Polarized Optical Microscopy (POM), Powder X-ray Diffraction (XRD), Vibrating Sample Magnetometry (VSM), Fluorescence Spectroscopy, H Nuclear Magnetic Resonance (NMR), UV/Vis Spectroscopy, Circular Dichroism (CD)	
Software: Microsoft Office, JMP, SolidWorks, Origin, Python, AI-assisted workflows (Claude)	
INTERESTS AND SKILLS	
Languages: Greek (Native), English (Fluent)	
Sports: Competitive and Semi-Professional level in soccer	
Music: piano, trumpet, guitar, and Byzantine music	